

A disproof of Conjecture 00000007665

The series $A_q(z)$ has no zeros on the negative half-axis

1 The conjecture

Conjecture 00000007665 reads:

Definition: The q -Airy function $A_q(z) = \sum (-1)^n q^{n(n-1)/2} z^n / (q; q)_n$ satisfies $A_q(qz) = 1 - zA_q(z)$. *Conjecture:* When q in $(0, 1)$ is algebraic, all zeros of $A_q(z)$ are real and simple, and the k -th zero on the negative half-axis satisfies $z_k = -q^{-k}(1 + O(q^{k/2}))$; furthermore the normalized limit of the adjacent spacing sequence follows an explicit density related to the tau function of the degenerate q -Painleve equation.

The Chinese version gives the same series with the summation range $n \geq 0$, and the same three clauses.

Reading. A_q is the function defined by the displayed series, with $(q; q)_n = \prod_{k=1}^n (1 - q^k)$ (so $(q; q)_0 = 1$). The conjecture is the conjunction, for every algebraic $q \in (0, 1)$, of

- (i) all zeros of A_q are real and simple;
- (ii) the zeros z_k of A_q on the negative half-axis satisfy $z_k = -q^{-k}(1 + O(q^{k/2}))$, i.e. there are C and K with $|z_k + q^{-k}| \leq C q^{-k} q^{k/2}$ for all $k \geq K$;
- (iii) a spacing law for the z_k (not specified precisely).

Clause (ii) presupposes that the k -th negative zero exists (for all large k); this is the reading we refute. Clause (iii) is not needed: the conjunction fails as soon as (ii) fails, and the Lean theorem is stated for an arbitrary proposition standing for (iii).

Remark on the “Definition” line. The functional equation stated there is not satisfied by the series: comparing coefficients of z^2 at $q = 1/2$ gives $q^2 a_2 = 1/3$ on the left and $-a_1 = 2$ on the right, where a_n is the n -th coefficient. No power series with constant term 1 and positive radius of convergence satisfies $A(qz) = 1 - zA(z)$ for $q \in (0, 1)$: the formal solution has coefficients $(-1)^n q^{-n(n+1)/2}$ and radius of convergence 0. So the series is the only meaningful definition, and we use it. This remark is not used below.

2 The result

Theorem 1. *Let $0 < q < 1$ and $x \leq 0$ be real. Then the series $A_q(x)$ converges and $A_q(x) \geq 1$. In particular A_q has no zero on the closed negative half-axis.*

Proof. Put $y = -x \geq 0$ and $t_n = q^{\binom{n}{2}} y^n / (q; q)_n$. Since $(-1)^n x^n = y^n$ and $n(n-1)/2 = \binom{n}{2}$, the n -th term of $A_q(x)$ equals t_n . Each factor $1 - q^k$ ($k \geq 1$) lies in $(0, 1]$, so $(q; q)_n > 0$ and $t_n \geq 0$. From $(q; q)_{n+1} = (q; q)_n (1 - q^{n+1})$ and $\binom{n+1}{2} = n + \binom{n}{2}$,

$$t_{n+1} = t_n \cdot \frac{q^n y}{1 - q^{n+1}}.$$

Because $q^n y \rightarrow 0$ and $1 - q^{n+1} \geq 1 - q > 0$, eventually $q^n y / (1 - q^{n+1}) \leq 1/2$, so $t_{n+1} \leq t_n/2$ for all large n and $\sum t_n$ converges (ratio test). All terms are nonnegative and $t_0 = 1$, hence $A_q(x) = \sum_n t_n \geq t_0 = 1 > 0$. $\square$

Corollary 2. *For every $q \in (0, 1)$ clause (ii) fails: there is no sequence $(z_k)_{k \geq K}$ of zeros of A_q on the negative half-axis, whatever bound they are asked to satisfy. Moreover, clause (i) together with clause (ii) read for arbitrary complex zeros (dropping the words “on the negative half-axis”) is also false. In particular the conjecture fails for the algebraic number $q = 1/2$.*

Proof. The first statement is immediate from Theorem 1. For the second, suppose all zeros are real and z_k are zeros with $|z_k + q^{-k}| \leq C q^{-k} (\sqrt{q})^k$ for $k \geq K$. Since $C(\sqrt{q})^k \rightarrow 0$, pick $k \geq K$ with $C(\sqrt{q})^k < 1$. Then $z_k = a$ is real and $a + q^{-k} \leq |a + q^{-k}| \leq q^{-k} \cdot C(\sqrt{q})^k < q^{-k}$, so $a < 0$, contradicting Theorem 1. The number $1/2$ is rational, hence algebraic. $\square$

Remark 3 (where the zeros are; not used in the proof). By Euler’s identity, quoted in the Wikipedia article “q-Pochhammer symbol” (https://en.wikipedia.org/wiki/Q-Pochhammer_symbol) as $(x; q)_\infty = \sum_{n=0}^{\infty} \frac{(-1)^n q^{n(n-1)/2}}{(q; q)_n} x^n$ with $(a; q)_\infty = \prod_{k=0}^{\infty} (1 - a q^k)$, the series is $A_q(z) = \prod_{k \geq 0} (1 - z q^k)$. Its zeros are $z = q^{-k}$ ($k \geq 0$): all real, simple and *positive*. So clause (i) is true, while clause (ii) fails because every zero lies on the positive half-axis. Only Theorem 1, which is proved directly from the series, is used and formalized.

3 Lean formalization

File `lean/Conjecture7665/Basic.lean` (Lean 4, Mathlib v4.33.1, only import `Mathlib`) contains, in namespace `C7665`:

- `qPoch q n =  $\prod_{k < n} (1 - q^{k+1})$` , and `qAiry q z =  $\sum'_n (-1)^n q^{n(n-1)/2} z^n / \text{qPoch } q n$` (complex `tsum`; $n(n-1)/2$ is natural division, which is exact). Convergence is proved, not assumed, where it is used.
- `AllZerosRealSimple q`: every zero z has $\text{Im } z = 0$ and `deriv (qAiry q) z  $\neq 0$` .
- `NegZerosAsymptotic q`: there are $z : \mathbb{N} \rightarrow \mathbb{R}$, C, K with, for $k \geq K$, $z_k < 0$, `qAiry q zk = 0` and $|z_k + (q^k)^{-1}| \leq C (q^k)^{-1} (\sqrt{q})^k$. `ZerosAsymptotic q` is the same with $z : \mathbb{N} \rightarrow \mathbb{C}$ and without $z_k < 0$.
- `term_summable` (ratio test), `one_le_tsum_term`, `qAiry_ofReal_of_nonpos`, and `qAiry_ne_zero_of_nonpos`: Theorem 1.
- `not_negZerosAsymptotic` and `not_allZerosReal_and_zerosAsymptotic`: Corollary 2 for every $q \in (0, 1)$.

Main theorems:

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theorem conjecture_7665_false (Spacing : R -> Prop) :
  not (forall q : R, IsAlgebraic Q q -> 0 < q -> q < 1 ->
    AllZerosRealSimple q /\ NegZerosAsymptotic q /\ Spacing q)
theorem conjecture_7665_false' (Spacing : R -> Prop) :
  not (forall q : R, IsAlgebraic Q q -> 0 < q -> q < 1 ->
    AllZerosRealSimple q /\ ZerosAsymptotic q /\ Spacing q)

```

(written here in ASCII; $R = \mathbb{R}$, $Q = \mathbb{Q}$). `lake build` succeeds and `#print axioms` reports, for both theorems, only `propext`, `Classical.choice`, `Quot.sound`; there is no `sorry` and no `native_decide`.

What is not refuted. A vacuous reading of clause (ii) (“for each k for which a k -th negative zero exists, ...”) is true, since there are none; under that reading clause (iii) concerns an empty sequence. Clause (i) is true (by the remark above). We refute the reading in which clause (ii) asserts negative zeros z_k for all large k , and the reading without “negative half-axis” combined with clause (i).